

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

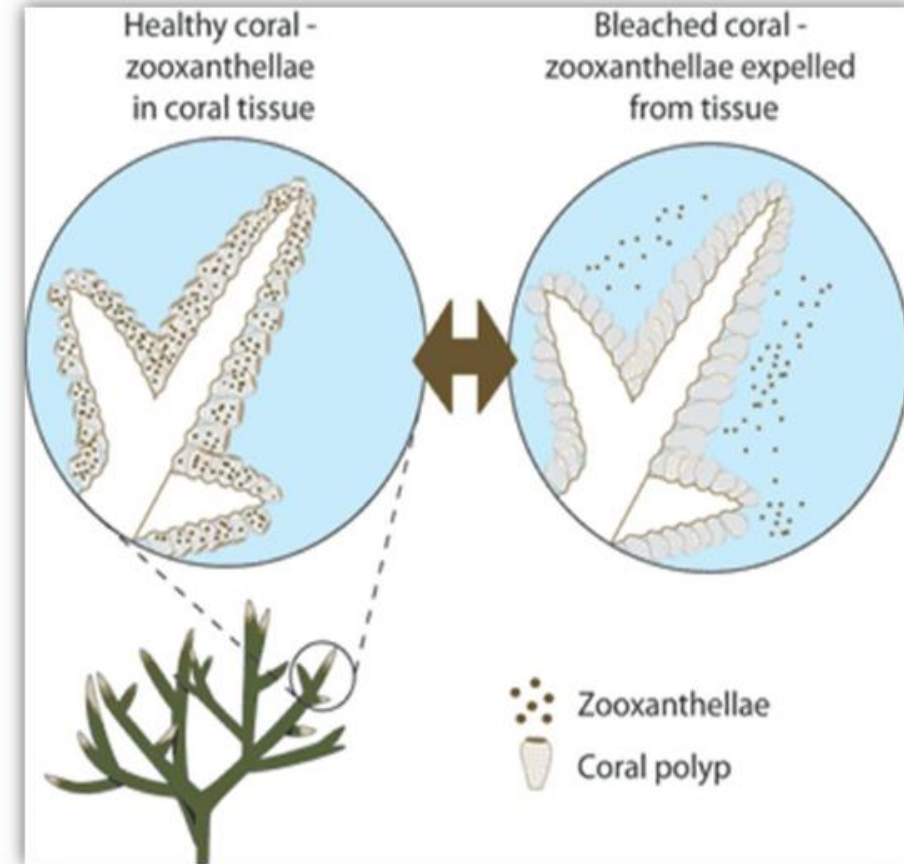
A COMPREHENSIVE COVERAGE

MODULE - 26

WATER POLLUTION - 3

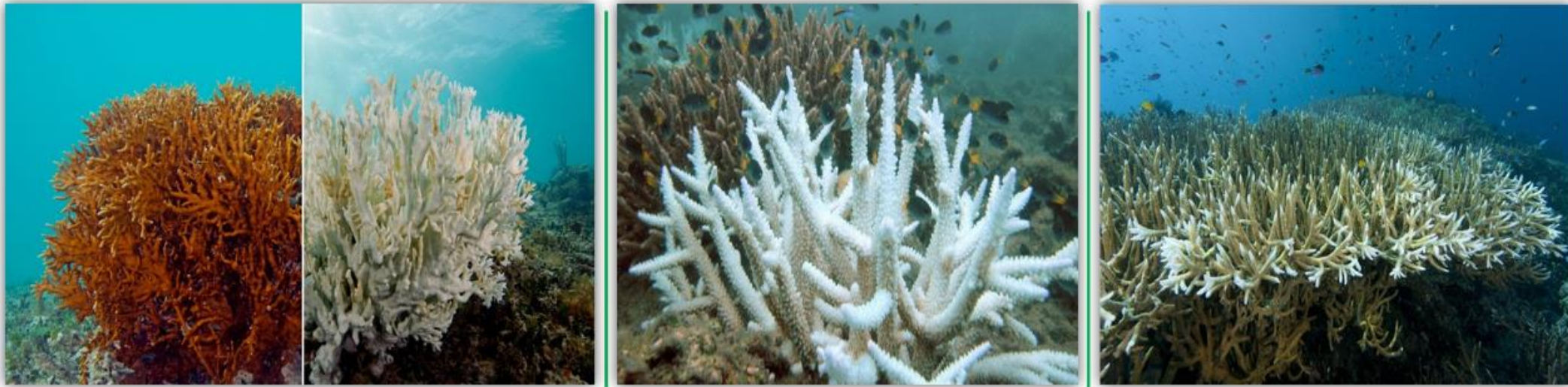
CORAL BLEACHING

- Coral bleaching is the stress response when the ideal conditions for coral growth are disturbed.
- When the ideal conditions for corals are disturbed, the symbiotic relationship is broken and zooxanthellae leaves the tissues of coral polyps. Then the polyps will lose their major source of energy and begin to starve (zooxanthellae provides coral polyps with up to 90% of its energy).
- The tissues of corals are actually not the beautiful colors of the coral reef, but are instead pale white. The corals receive their coloration from the Zooxanthellae living within their tissues.



CORAL BLEACHING

- When zooxanthellae leave the coral polyps, they revert back to their original color which is pale white. This phenomenon is known as coral bleaching.
- When corals bleach, they commonly lose 60-90% of their zooxanthellae. If the stress-causing factor is not too severe and if it decreases in time, the affected corals usually regain their symbiotic algae within several weeks or a few months.
- If zooxanthellae loss is prolonged, (if the stress continues) then coral polyps eventually die. So coral bleaching and coral death are not the same events.



CORAL BLEACHING

CAUSES OF CORAL BLEACHING

- 1. Temperature Anomalies** - Coral polyps have a narrow temperature tolerance range, they cannot withstand very low or very high temperatures. They thrive in warm waters. So abnormal high or low temperatures can induce coral bleaching.
- 2. Ocean Acidification** - Increased acidity of ocean water also causes coral bleaching. Ocean acidification has reduced corals calcifying ability.
- 3. Ultraviolet (UV) radiation** - Coral Polyps are very sensitive organisms. Exposure to UV rays causes coral bleaching and death of coral polyps.
- 4. Sudden exposure** of coral reefs to the atmosphere during events such as extreme low tides can cause bleaching.
- 5. Dumping of Pollutants** from Industries.



CORAL BLEACHING

IMPORTANCE OF CORAL REEFS

- Coral reefs support some of the most bio-diverse ecosystems on the planet. Thousands of marine species depend on coral reefs for survival, including species of sea turtles, fish, crabs, shrimp, jellyfish, sea birds, starfish and more.
- Coral reefs provide shelter, nesting grounds, and protection from predators.
- Coral bleaching impacts peoples' livelihoods and food security.
- Bleached coral reef aggravates the overfishing crisis by removing links in the food web and depriving some fish and crustacean species of a place to lay eggs.
- If global warming remains on its upward path, by 2050 just 5% of Australia's Great Barrier Reef, (the world's largest coral reef) will remain. It's not only a tragedy for wild life, around half a billion people rely on fish from coral reefs as their main source of protein.
- Finally, reef tourism brings in billions of dollars each year and supports thousands of jobs. Bleached coral reefs, devoid of marine species, jeopardize it.



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